Experiment 15: Operating System Detection Using Nmap in Ubuntu

1. Aim

To identify the operating system of a target computer using the Nmap tool in Ubuntu.

2. Requirements

- Ubuntu operating system
- Nmap installed
- Internet or local network connection (depending on the target)
- Permission to scan the target computer

3. Procedure

Step 1: Open Terminal

Press Ctrl + Alt + T to open the Ubuntu Terminal.

Step 2: Check whether Nmap is installed

Type:

nmap --version

If Nmap is installed, its version information will appear.

If it is not installed, run:

sudo apt update

sudo apt install nmap

Step 3: Identify your own computer's IP address

ip addr

Look for the IP address assigned to your active network interface. For a simple first test, you can scan your own computer using the loopback address 127.0.0.1.

Step 4: Perform OS detection

Run the following command to scan your own computer:

sudo nmap -O 127.0.0.1

- sudo — runs Nmap with administrator privileges.
- nmap — starts the network scanning tool.
- -O — enables operating system detection. This is the capital letter O, not zero.
- 127.0.0.1 — refers to your own computer.

In Windows:

Installing nmap in windows:

winget install Insecure. Nmap

nmap --version

nmap <IP ADDRESS>